



Working in the MXCuBE collaboration



Collaborating since 2005



Introduction and Working in the project

Beamline control and software development



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(Alternance)

Scientific coordination



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- The aim is to provide a platform for sharing solutions and know-how:
- We are striving towards making MXCuBE and easy to extend, deploy and use application
- We aim to keep the user experience and look and feel consistent across sites.
- During my years in the project; a very friendly and collaborative spirit with a solution oriented mindset.
- The code base is over 20 years old, there is quite alot of legacy code.



It's a collaborative effort,
A very big thanks to all of you !



Organisation

- Steering committee
 - Comprised of one representative per site
 - Meeting every MXCuBE meeting, two times every year
- Scientific committee
 - Meets with the developers twice a year during the MXCuBE meeting
- Developers committee
 - Monthly developers meetings

- Like in most projects there are some conventions and good practices
- We try to base those on what's widely used in the software community
- Document in the CONTRIBUTING.md file
<https://github.com/mxcube/mxcubecore/blob/develop/CONTRIBUTING.md>

Brief summary

- Bugs, use the GitHub issues. Check for duplicates and provide as much information as possible
- Docstring are written with google style doc-strings
 - <https://mxcube.web.readthedocs.io/en/stable/>
 - <https://mxcube.core.readthedocs.io/en/stable/>
- Ruff, with rules: "E", "F", "B", "I", "S" with some global exceptions

Recommendation

- Consider using an editor with good support for formatting, linting and testing
- vscode, pycharm, (emacs or vi if one still is adventurous)
- Use conda or similar tool to handle your virtual environment and to install development dependencies using:
 - poetry install - -with dev
- Consider using pre-commit hook

Github actions based CI

- PyTest
- Lint and format
- Tag and publish package to PyPi
- Build of documentation
 - <https://mxcube.web.readthedocs.io/en/latest/>
 - <https://mxcube.core.readthedocs.io/en/latest/>
- Dependabot
- GitHub advanced security with CodeQL

Pull requests

- Pull requests are peer reviewed
- To be merged a PR needs a review from another developer (preferably from another institute) *within 15 days*
- We welcome discussions on PRs. We believe it is an excellent way to learn about MXCuBE and the technologies used
- The feedback on a PR should be kept constructive, informative and preferably with some context that explains the comment

Pull requests

- To help our colleagues to review, keep the PRs as small as possible. This also makes the PR easier to manage and track the change if needed
- Explain what the PR does and explain the details that may need explaining with a comment
- We understand that it's sometimes difficult to keep PR's small, occasional exceptions are acceptable especially if they are for site specific code

Thank you for your attention

